CashALL Partner Android App — Complete Technical Handoff

EXHAUSTIVE BACKEND INTEGRATION SPECIFICATION & REAL PRODUCTION PLAYBOOK

Target System: CashALL Production Re-Commerce Engine	**Repository:** github.com/cashall7003216788-jpg/CashALL
Production API URL: https://www.cashall.in/api/v1	**Architecture:** Next.js 14 App Router, Supabase Postgres, Prisma
Target Audience: Partner Android App Developer & Backend Team	**Document Scope:** End-to-End Specs, Payloads, Routing, Roadmap

CRITICAL ARCHITECTURAL CLARIFICATION: WHY https://cashall.in/api/v1 RETURNS 404 IN BROWSER

Developer Observation: *'I opened https://cashall.in/api/v1 in Chrome and got 404 This page could not be found. Is the backend down or missing?'*

Authoritative Answer: NO. The backend is 100% ONLINE, CONNECTED TO POSTGRESQL, AND OPERATIONAL.
• **Next.js App Router Mechanism:** In Next.js 14, routing is strictly mapped to individual files (route.ts). In our repository, app/api/v1/ is a directory containing specialized sub-routes; there is no index app/api/v1/route.ts file. Therefore, navigating to the folder root /api/v1 in a browser will return a 404 error page by design.
• **Live Verification:** Navigating to https://www.cashall.in/api/health returns HTTP 200: {"status":"ok", "timestamp":"2026-09-01T...", "service":"CashALL API", "environment":"production", "database":"connected"}.
• **Action for Android Developer:** Do not ping the root /api/v1 in a browser. Your Android networking client (Retrofit/Ktor) must invoke the specific sub-path endpoints (e.g. POST /api/v1/agent/login, GET /api/v1/agent/orders, POST /api/v1/quotes/calculate).

1. Developer Queries Resolution & Backend Roadmap

Question by Developer: *'Are we planning to add these APIs in backend? (No PricingRule endpoint, No image-upload endpoint, No IMEI blocklist, no app-settings endpoint, No agent-side cancel, no arrival event, No Idempotency-Key support)'*
Authoritative Answer: YES. Here is the exact status of each item, how to connect today, and our backend deployment roadmap:

Developer Query	Current Codebase Status	How Android App Connects Today	Backend Plan & Roadmap
1. No PricingRule endpoint	Pricing rules reside in PostgreSQL PricingRule table. The backend does NOT expect mobile app to download raw rules.	Call POST /api/v1/quotes/calculate with variantId and answers array. Server returns exact valuation.	**NO CODE DUPLICATION.** Server executes calculation. If offline mode is required, backend will add GET /catalog/pricing-rules.
2. No image-upload endpoint	**ALREADY EXISTS!** The developer was unaware: TWO active upload endpoints exist backed by Supabase Storage.	**A:** POST /api/v1/storage/upload (QC photos). **B:** POST /api/v1/agent/upload-payment (UPI proofs).	Fully supported today. Both accept multipart/form-data and return signed Supabase URLs.
3. No IMEI blocklist & app-settings	ImeiService.verifyIMEI exists in backend but is not linked to active agent route. SystemSetting exists in DB.	Currently, IMEI is validated during inspection submission in /agent/orders/[id]/inspection.	**PLANNING TO ADD:** • POST /api/v1/imei/verify • GET /api/v1/settings/app (min app version, support phone).
4. No agent-side cancel & arrival event	OrderStatus enum has PARTNER_ON_THE_WAY & CANCELLED. Generic POST /orders/[id]/cancel exists.	For cancellations today, mobile app calls POST /api/v1/orders/[id]/cancel with {"reason": "..."}.	**PLANNING TO ADD:** • POST /agent/orders/[id]/pickup-failed • PATCH /agent/orders/[id]/status (arrival events).
5. No Idempotency-Key support	Backend currently does not inspect Idempotency-Key header on payment mutations.	Android App must start sending Idempotency-Key: <UUIDv4> immediately and disable button on tap.	**PLANNING TO ADD:** Server-side idempotency middleware caching responses by key in PostgreSQL.

2. Complete Active API Inventory for Partner App

Below is the complete inventory of live production APIs that the Android developer should connect with:

Method	Endpoint	Purpose	Auth	Caller	Status	Mobile Need
`POST`	`/api/v1/agent/login`	Field agent login with phone & password	None	Agent	EXISTS	**CRITICAL**
`GET`	`/api/v1/agent/orders`	Fetch assigned doorstep orders	Query param	Agent	EXISTS	**CRITICAL**
`POST`	`/api/v1/agent/orders/[id]/inspection`	Submit physical inspection & set offer	None	Agent	EXISTS	**CRITICAL**
`POST`	`/api/v1/agent/upload-payment`	Upload UPI screenshot & 12-digit UTR	None	Agent	EXISTS	**CRITICAL**
`POST`	`/api/v1/agent/orders/[id]/complete`	Complete order, sync Sheets & email bill	None	Agent	EXISTS	**CRITICAL**
`POST`	`/api/v1/storage/upload`	Upload QC device inspection photos	Bearer Token	User/Agent	EXISTS	**HIGH**
`POST`	`/api/v1/quotes/calculate`	Run backend pricing engine algorithm	None	Public/Agent	EXISTS	**CRITICAL**

GET/POST	`/api/v1/orders/[id]/generate-bill`	Retrieve structured legal bill / PDF data	None	Public/Agent	**EXISTS**	**HIGH**
GET	`/api/v1/catalog/questions`	Fetch condition questions & options matrix	None	Public/Agent	**EXISTS**	**HIGH**
POST	`/api/v1/orders/[id]/cancel`	Cancel order / unassigned pickup	None	Public/Agent	**EXISTS**	**HIGH**

3. Detailed API Contracts & Copy-Paste Technical Specifications

Below are the complete, exact HTTP contracts for every endpoint the Android developer must call, including headers, curl examples, JSON requests, and response payloads:

3.1 Agent Authentication: POST /api/v1/agent/login

Headers: Content-Type: application/json
Curl Example:
curl -X POST https://www.cashall.in/api/v1/agent/login -H 'Content-Type: application/json' -d '{"name":"SANGEET SHAW", "password":"6289477287"}'
Request Body:
{"name": "SANGEET SHAW", "password": "6289477287"}
Success Response (HTTP 200):
{"success": true, "token": "tok_agent_sangeetshaw_1725261890", "agent": {"id": "agent_sangeet_shaw", "name": "SANGEET SHAW", "email": "sangeetshaw39@gmail.com", "phone": "6289477287", "role": "AGENT"}}
Android Handling: Store token in EncryptedSharedPreferences. Transmit as Authorization: Bearer <token> on subsequent requests.

3.2 Fetch Assigned Orders: GET /api/v1/agent/orders

Query Params: agentId or phone or name
Headers: Authorization: Bearer <token>
Curl Example:
curl -X GET 'https://www.cashall.in/api/v1/agent/orders?phone=6289477287' -H 'Authorization: Bearer <token>'
Success Response (HTTP 200):
{"success": true, "orders": [{"id": "b1a2c3d4-e5f6-4a5b-8c9d-0e1f2a3b4c5d", "orderNumber": "CA36738", "customerName": "Rahul Sharma", "customerPhone": "9876543210", "customerEmail": "rahul@gmail.com", "deviceName": "Apple iPhone 13 (128 GB)", "estimatedPrice": 28000, "finalPrice": null, "address": "Flat 402, Green Heights, Salt Lake, Kolkata - 700091", "pickupDate": "Today", "pickupTimeSlot": "2:00 PM - 4:00 PM", "status": "PARTNER_ASSIGNED", "paymentStatus": "PENDING", "selectedAnswersJson": "{\"screen\":\"flawless\",\"body\":\"minor_scratches\"}", "breakdownJson": "[...]"}]}

3.3 Submit Inspection Findings: POST /api/v1/agent/orders/[id]/inspection

Path Param: id = Order UUID or orderNumber (e.g. CA36738)
Headers: Content-Type: application/json, Authorization: Bearer <token>
Request Body:
{"imei": "356938114059123", "screenFinding": "minor_scratches", "bodyFinding": "flawless", "inspectedAnswers": {"q-warranty": "o-w-no", "q-scratches": "o-sc-1-2"}, "revisedPrice": 26500, "reason": "Screen has hairline scratches", "customerEmail": "rahul@gmail.com", "agentName": "SANGEET SHAW"}
Success Response (HTTP 200):
{"success": true, "message": "Physical inspection completed! Final payout offer locked at ■26,500.", "order": {"id": "b1a2c3d4-...", "orderNumber": "CA36738", "status": "ACCEPTED", "finalPrice": 26500}}

3.4 Dynamic Pricing Valuation: POST /api/v1/quotes/calculate

Headers: Content-Type: application/json
Request Body:
{"variantId": "v-iphone-13-128", "answers": [{"questionId": "q-screen-defect", "questionTitle": "Screen Defect", "group": "SCREEN", "optionId": "o-s-cracked", "optionLabel": "Cracked Glass"}]}
Success Response (HTTP 200):
{"success": true, "data": {"basePrice": 28000, "totalDeductions": 6000, "totalBonuses": 0, "estimatedPrice": 22000, "breakdown": [{"category": "SCREEN", "title": "Screen Defect", "selection": "Cracked Glass", "amount": -6000}]}}
Note: Call this endpoint whenever the agent alters defect questions at the doorstep. Never code pricing formulas inside Android.

3.5 Upload QC Inspection Photos: POST /api/v1/storage/upload

Headers: Content-Type: multipart/form-data, Authorization: Bearer <token>
Form Data Fields:
• file: Binary file (JPEG/PNG/WebP, max 5MB)
• bucket: qc-reports
• orderId: b1a2c3d4-e5f6-... (Order UUID)
Success Response (HTTP 200):
{"success": true, "data": {"path": "agent_id/unique-uuid.jpg", "url": "https://jqysknhobtpcbyyltnfc.supabase.co/storage/v1/object/sign/qc-reports/..."}}

3.6 Upload UPI Payment Proof: POST /api/v1/agent/upload-payment

Headers: Content-Type: multipart/form-data, Idempotency-Key: <UUIDv4>
Form Data Fields:
• file: Binary image (UPI payment confirmation screenshot)
• orderId: CA36738 (or Order UUID)
• urn: 424918274012 (12-digit bank UTR reference number)
• agentId: agent_sangeet_shaw
Success Response (HTTP 200):
{"success": true, "message": "Payment screenshot & 12-digit URN uploaded and saved.", "urn": "424918274012", "paymentScreenshotUrl": "https://jqysknhobtpcbyyltnfc.supabase.co/storage/v1/object/sign/payment-screenshots/..."}

3.7 Complete Order & Email Bill: POST /api/v1/agent/orders/[id]/complete

Path Param: id = Order UUID or orderNumber (e.g. CA36738)
Headers: Content-Type: application/json, Idempotency-Key: <UUIDv4>
Request Body:
{"finalPrice": 26500, "utr": "424918274012", "agentName": "SANGEET SHAW"}
Success Response (HTTP 200):
{"success": true, "message": "Order #CA36738 marked COMPLETED. Tax invoice PDF emailed to rahul@gmail.com.", "order": {"orderNumber": "CA36738", "status": "COMPLETED", "finalPrice": 26500, "urn": "424918274012"}}
Server Automation: Automatically sets Order status to COMPLETED, marks Payment as PAID, writes row to Google Sheets, and emails official invoice.

3.8 Retrieve Legal Purchase Receipt / Bill: GET /api/v1/orders/[id]/generate-bill

Path Param: id = Order UUID or orderNumber (e.g. CA36738)
Success Response (HTTP 200):
{"success": true, "data": {"billData": {"billNumber": "CA36738_2026", "orderNumber": "CA36738", "seller": {"name": "Rahul Sharma", "phone": "9876543210", "address": "Kolkata, WB"}, "buyer": {"name": "AARNA ENTERPRISE", "platform": "CashALL Platform", "gstin": "19AVPPG9800JIZ3", "address": "Howrah, West Bengal"}, "device": {"deviceName": "Apple iPhone 13 (128 GB)", "imei1": "356938114059123"}, "financials": {"finalPurchasePrice": 26500, "utrNumber": "424918274012", "paymentStatus": "PAID"}, "declarations": {"sellerDeclarationText": "This is a computer-generated receipt issued by CashALL. The seller has voluntarily sold the device at the agreed final price..."}}}}
Use for on-screen receipt display or printing via Bluetooth thermal printer.

3.9 Order Cancellation: POST /api/v1/orders/[id]/cancel

Path Param: id = Order UUID or orderNumber (e.g. CA36738)
Headers: Content-Type: application/json
Request Body: {"reason": "Customer refused doorstep price offer / not available"}
Success Response (HTTP 200): {"success": true, "message": "Order successfully cancelled.", "data": {"order": {"status": "CANCELLED"}}}

4. Single Source of Truth: Pricing Engine Architecture

CRITICAL DIRECTIVE: DO NOT CODE INDEPENDENT VALUATION LOGIC IN ANDROID.

The Partner App must calculate identical prices to the CashALL website. Both must share the backend PostgreSQL pricing rules.

1. Where Pricing Logic Lives:

- **Database Table:** PricingRule in Supabase PostgreSQL (AWS Mumbai).
- **Backend Service:** lib/services/pricing.service.ts -> PricingService.calculateQuote(variantId, answers).
- **Live Endpoint:** POST https://www.cashall.in/api/v1/quotes/calculate.

2. Mathematical Rules Executed by the Engine:

- **Base Price:** Retrieved from DeviceVariant.basePrice.
- **Percentage Deductions:** deduction = -(basePrice * rule.adjustmentValue) / 100.
- **Fixed Deductions:** deduction = -rule.adjustmentValue.
- **Fixed Bonuses:** bonus = +rule.adjustmentValue.
- **Floor Price Rule:** finalPrice = Math.max(500, Math.round(basePrice * 0.15), rawEstimated).

3. Mobile App Integration Pattern:

When the field agent toggles a screen crack, body dent, or missing box at the doorstep, the Android app makes a lightweight HTTP POST to /api/v1/quotes/calculate and displays the returned estimatedPrice and breakdown. This guarantees 100% price parity with the customer's web quote.

5. Supabase Architecture & Mobile Access Rules

CRITICAL SECURITY RULE: The Partner Android App must communicate SOLELY via CashALL Next.js API routes (Option A). The mobile app MUST NEVER directly connect to Supabase PostgreSQL or use the Supabase Service Role Key.

Configuration / Credential Name	Classification	Mobile App Handling Rule
`NEXT_PUBLIC_SUPABASE_URL`	**PUBLIC CLIENT CONFIG**	Optional. Only needed if client-side direct SDK used; otherwise NOT required.
`NEXT_PUBLIC_SUPABASE_ANON_KEY`	**PUBLIC CLIENT CONFIG**	Restricted by Row Level Security. Only provide if mobile uses Supabase client.
`SUPABASE_SERVICE_ROLE_KEY`	**SERVER SECRET**	**NEVER PROVIDE TO DEVELOPER. NEVER BUNDLE IN APK.** Full DB admin bypass.
`DATABASE_URL / POSTGRES_PRISMA_URL`	**SERVER SECRET**	**NEVER PROVIDE TO DEVELOPER.** Direct PostgreSQL pool connection string.
`SUPABASE_JWT_SECRET`	**SERVER SECRET**	**NEVER PROVIDE TO DEVELOPER.** Used exclusively by server to sign/verify JWTs.
`Database Password`	**SERVER SECRET**	**NEVER PROVIDE TO DEVELOPER.** PostgreSQL superuser credential.

6. Vercel Requirements & Staging Environment

Vercel Access:

"Partner App developer only needs the production API URL; Vercel access is not required."

The developer compiles native Android code. Serverless routing and backend deployments are managed by the CashALL engineering team.

Staging Environment Status:

"No separate staging environment is currently configured."

The project operates on a single production branch (main) with one PostgreSQL database. During mobile development, the developer must test against https://www.cashall.in/api/v1 using dedicated test orders prefixed #TEST_... assigned to the test agent account (SANGEET SHAW) to avoid polluting production accounting.

7. Exact Access Checklist: What to Give vs What NOT to Give

Category	Resource / Credential Item	Action	Technical & Security Reason
Repository	**GitHub Access (github.com/cashall7003216788-jpg/CashALL)**	GIVE	Allows developer to review API request/response models and schemas.
API URL	**Production API URL: https://www.cashall.in/api/v1**	GIVE	Base URL for Android HTTP Retrofit/Ktor networking client.
Health Check	**Server Health URL: https://www.cashall.in/api/health**	GIVE	Used by mobile app for connectivity and network reachability checks.
Test Accounts	**Test Agent Account (SANGEET SHAW / 6289477287)**	GIVE	Allows testing login, order listing, inspection, and completion flows.
Test Orders	**Synthetic Test Orders prefixed TEST_...**	GIVE	Prevents corrupting live customer accounting during development testing.
Push Services	**Firebase google-services.json (Client-only)**	GIVE (If FCM)	Required for FCM push notifications in Android Studio. Contains only public app IDs.
Database	**DATABASE_URL & PostgreSQL Password**	DO NOT GIVE	Direct DB connection strings allow bypassing backend business logic.
Supabase Key	**SUPABASE_SERVICE_ROLE_KEY**	DO NOT GIVE	Admin bypass key for Supabase RLS and storage. Server secret.
JWT Secret	**SUPABASE_JWT_SECRET**	DO NOT GIVE	Private secret used by backend server to generate and verify signatures.
Hosting	**Vercel Project Access or Personal Access Token**	DO NOT GIVE	Developer does not manage production cloud deployments.
Admin Operator	**Admin Portal Credentials (ADMIN_PASSWORD)**	DO NOT GIVE	Partner developer must never possess administrator portal credentials.

8. Executive One-Page Developer Handover Summary

Copy and paste this final checklist directly to the Android developer:

CHECKLIST: ITEMS TO HAND OVER TO PARTNER APP DEVELOPER
[ ] 1. GitHub Repository: https://github.com/cashall7003216788-jpg/CashALL (Grant Read access).
[ ] 2. Production Base URL: https://www.cashall.in/api/v1 (Call specific endpoint sub-paths; do NOT ping /api/v1 root).
[ ] 3. Server Health Check URL: https://www.cashall.in/api/health (Returns {"status":"ok", "database":"connected"}).
[ ] 4. Test Agent Credentials: • **Name:** SANGEET SHAW • **Phone / Password:** 6289477287 (Master Password: Ank933967@) • **Role:** AGENT
[ ] 5. Pricing Engine: Call POST /api/v1/quotes/calculate with defect options. Do not write duplicate pricing formulas in Android.
[ ] 6. Image Uploads: Use POST /api/v1/storage/upload (for QC photos) & POST /api/v1/agent/upload-payment (for UPI proofs).
[ ] 7. Idempotency Header: Transmit Idempotency-Key: <UUIDv4> on payment mutations right now.
[ ] 8. Backend Roadmap: Backend team is deploying pickup-failed, arrival events (status), and settings/app in the current sprint.
[ ] 9. PDF Technical Spec: Provide this complete file CashALL_Partner_App_Backend_Access_and_API_Handoff.pdf.